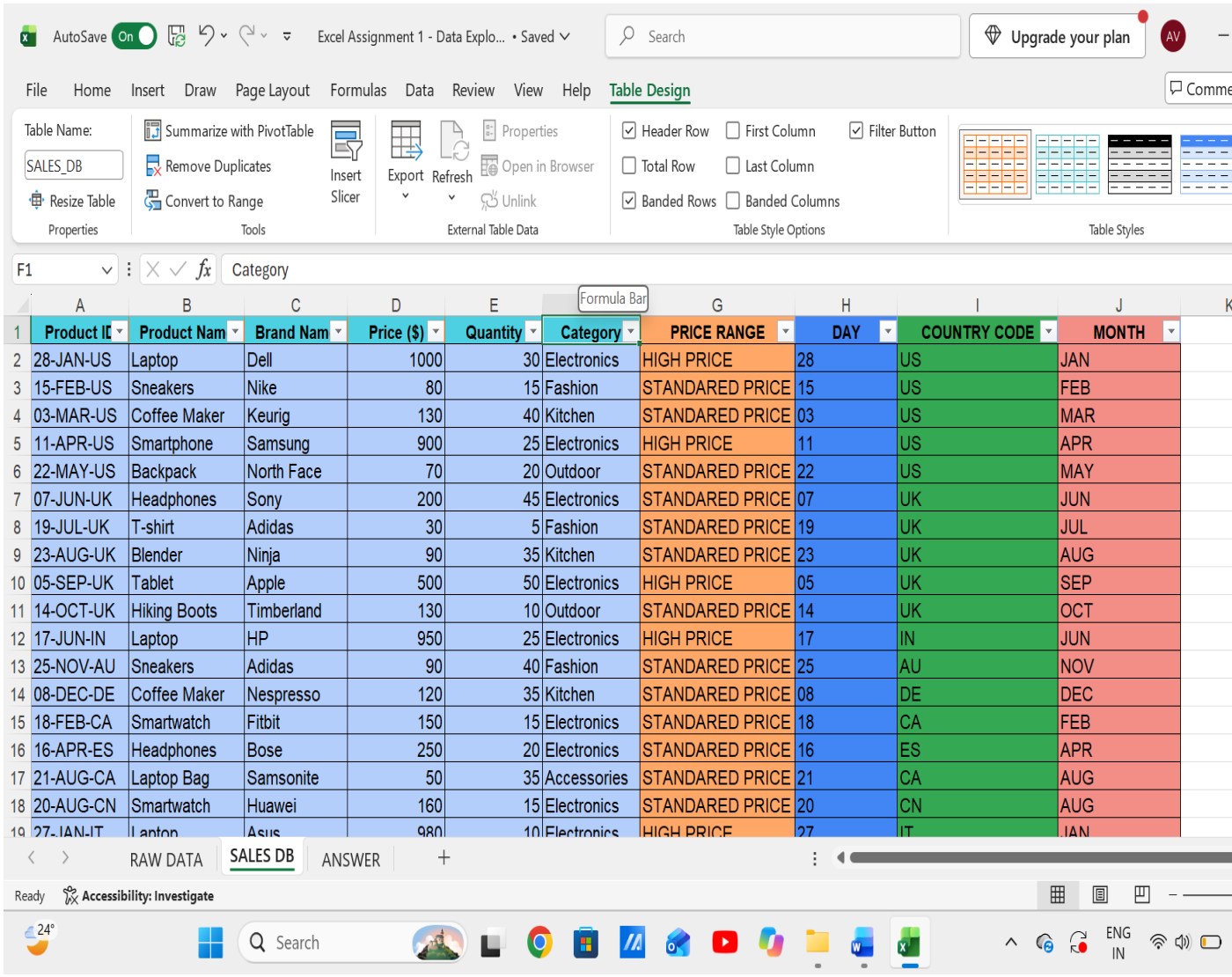
BASIC DATA EXPLORATION

STEP – 1

Converting raw dataset into table and given Table Name **“Sales_DB”**



Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category	PRICE RANGE	DAY	COUNTRY CODE	MONTH
28-JAN-US	Laptop	Dell	1000	30	Electronics	HIGH PRICE	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	STANDARED PRICE	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	STANDARED PRICE	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	HIGH PRICE	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	STANDARED PRICE	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	STANDARED PRICE	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	STANDARED PRICE	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	STANDARED PRICE	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	HIGH PRICE	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	STANDARED PRICE	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	HIGH PRICE	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	STANDARED PRICE	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	STANDARED PRICE	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	STANDARED PRICE	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	STANDARED PRICE	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	STANDARED PRICE	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	STANDARED PRICE	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	HIGH PRICE	27	IT	JAN

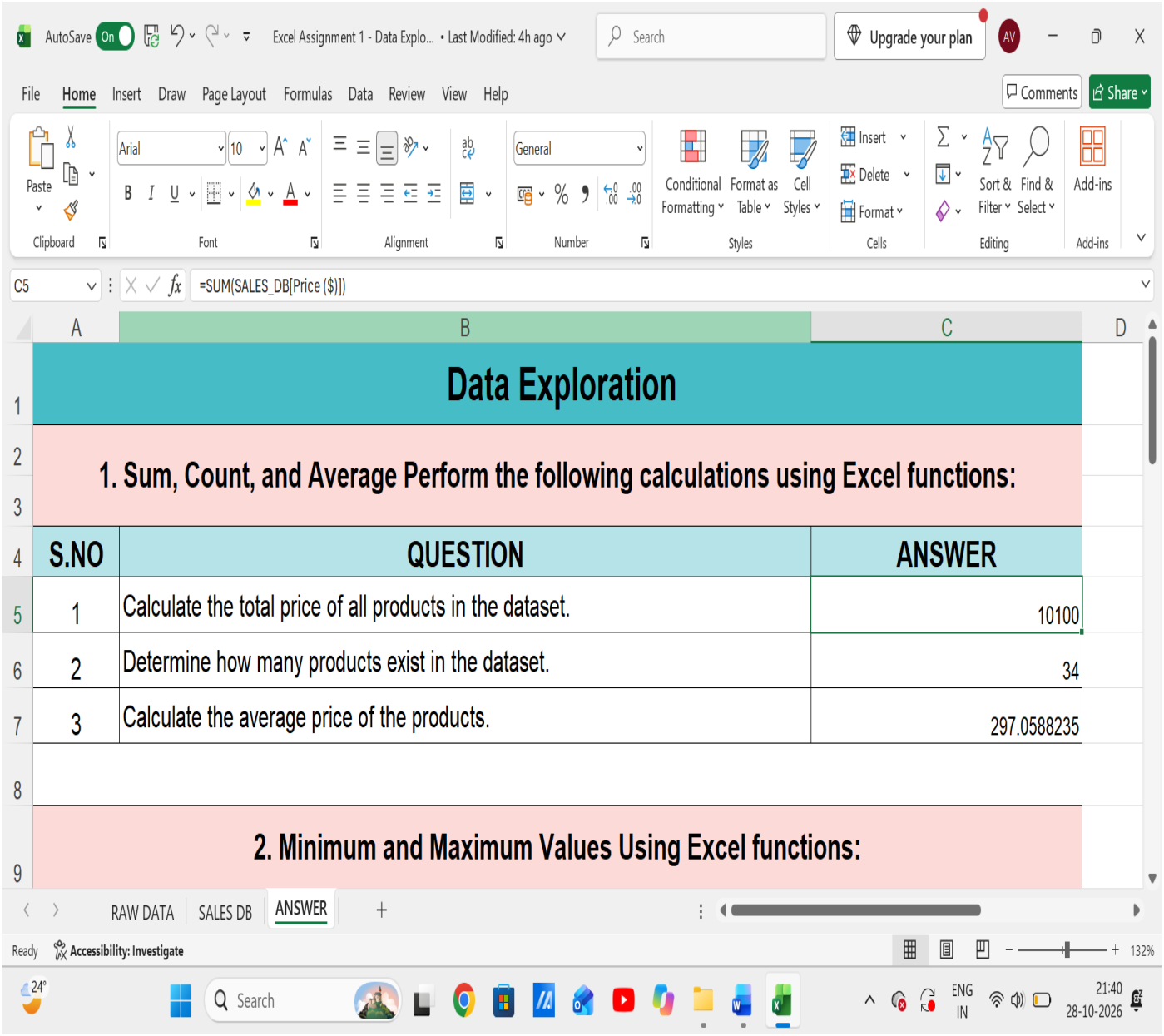
STEP – 2

Sum, Count, and Average:

Perform the following calculations using Excel functions:

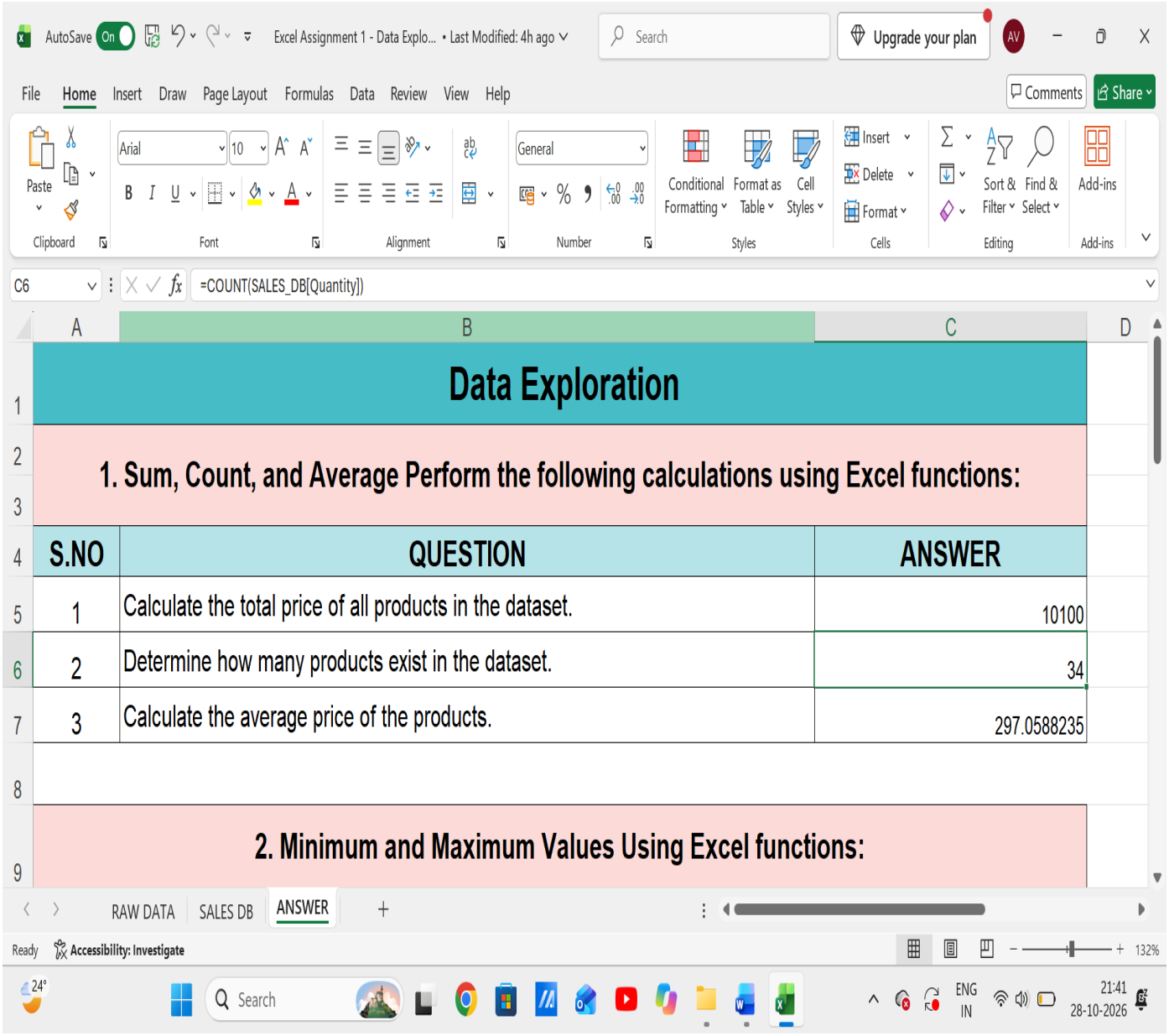
1. Calculate the total price of all products in the dataset.

Here using **Sum () Function**



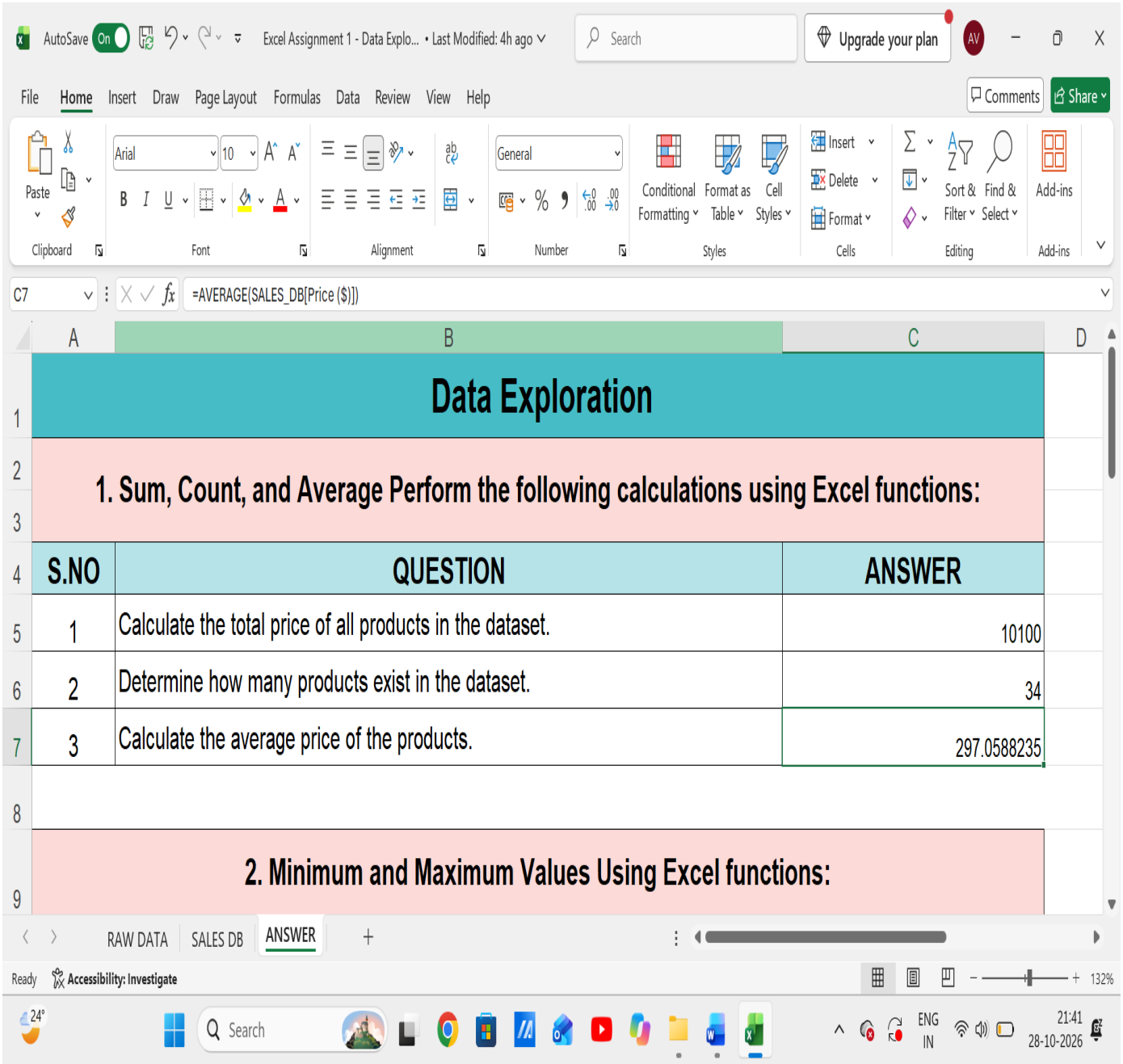
2.Determine how many products exist in the dataset.

Here using **Count () Function**



3.Calculate the average price of the products.

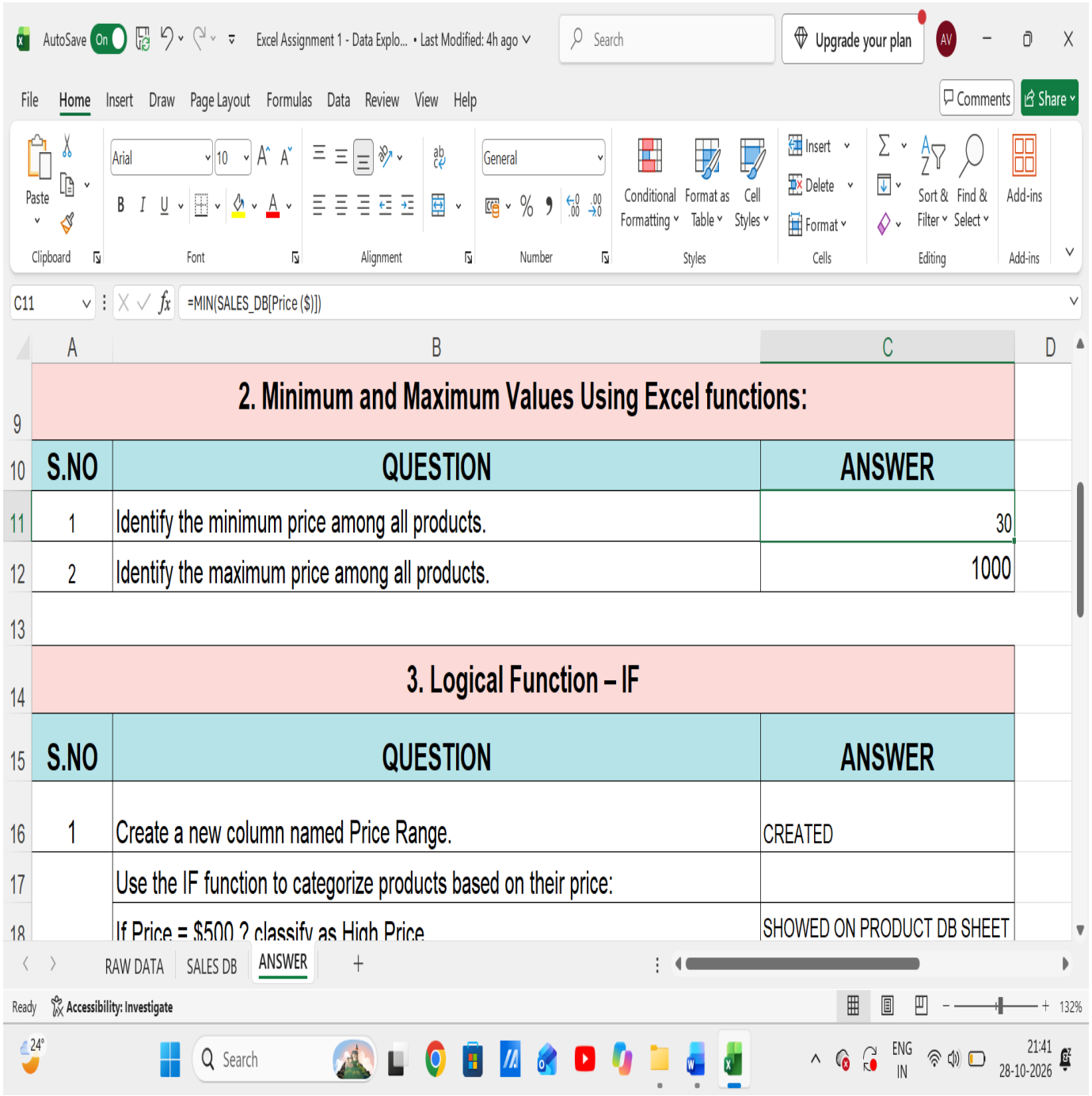
Here using **Average () function**



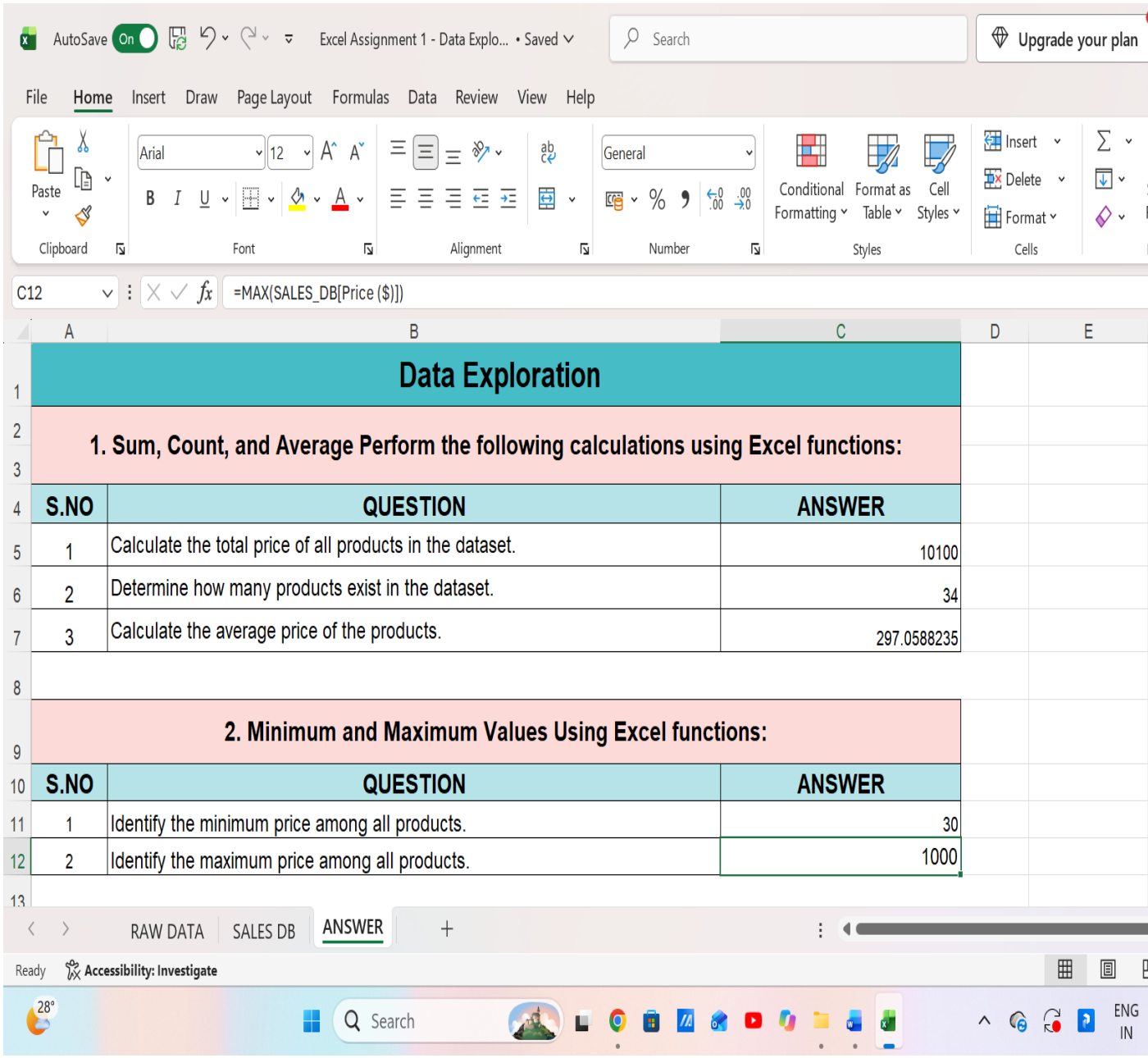
STEP 3

Minimum and Maximum Values Using Excel functions:

1.Identify the minimum price among all products.



2.Identify the maximum price among all products.



STEP-4

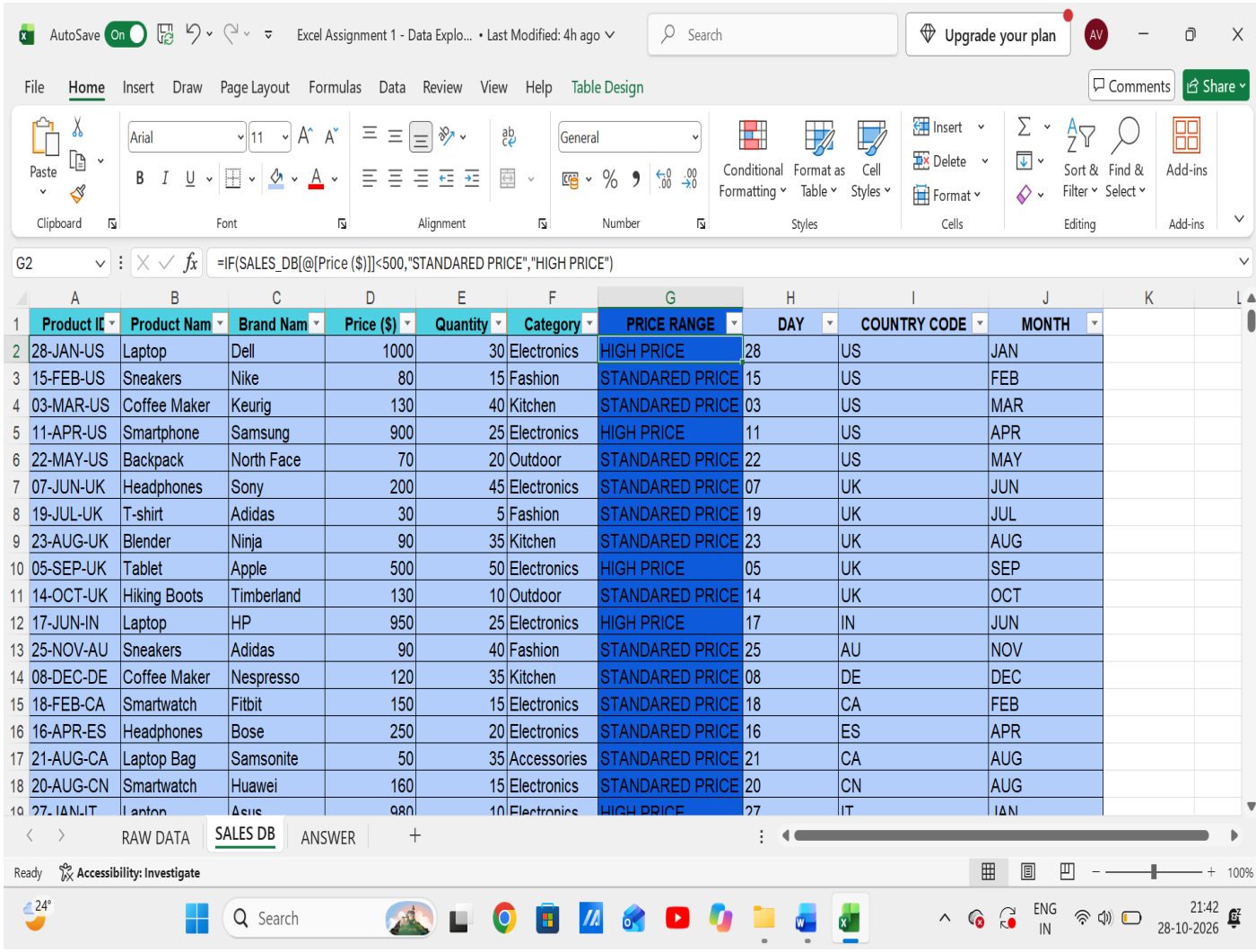
Logical Function – IF

1.Create a new column named Price Range.

2. Use the IF function to categorize products based on their price:

I. If Price = $500 → classify as **High Price**

II. If Price < $500 → classify as **Standard Price**



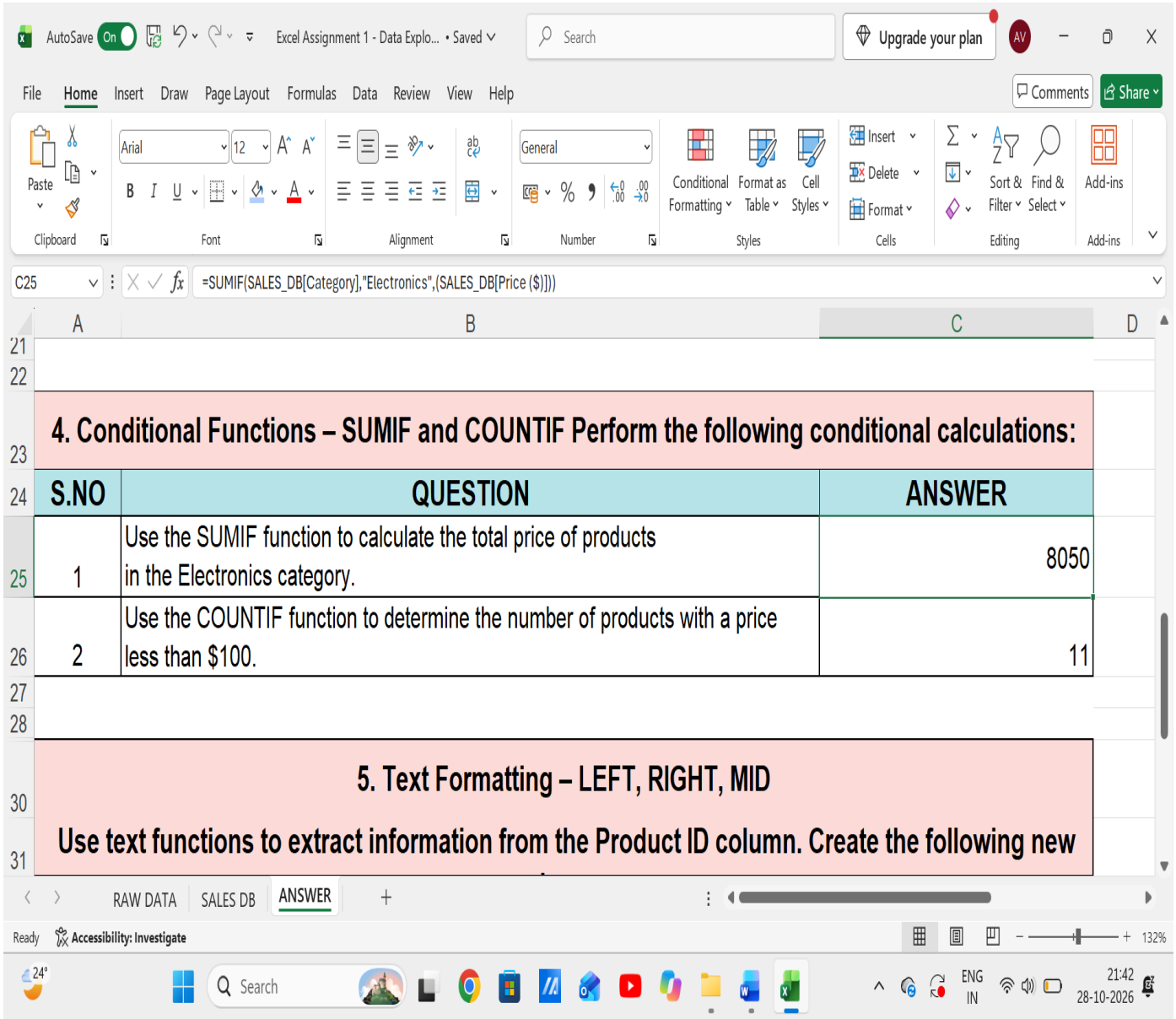
	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category	PRICE RANGE	DAY	COUNTRY CODE	MONTH
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	HIGH PRICE	28	US	JAN
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	STANDARED PRICE	15	US	FEB
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	STANDARED PRICE	03	US	MAR
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	HIGH PRICE	11	US	APR
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	STANDARED PRICE	22	US	MAY
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	STANDARED PRICE	07	UK	JUN
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	STANDARED PRICE	19	UK	JUL
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	STANDARED PRICE	23	UK	AUG
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	HIGH PRICE	05	UK	SEP
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	STANDARED PRICE	14	UK	OCT
12	17-JUN-IN	Laptop	HP	950	25	Electronics	HIGH PRICE	17	IN	JUN
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	STANDARED PRICE	25	AU	NOV
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	STANDARED PRICE	08	DE	DEC
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	STANDARED PRICE	18	CA	FEB
16	16-APR-ES	Headphones	Bose	250	20	Electronics	STANDARED PRICE	16	ES	APR
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	STANDARED PRICE	21	CA	AUG
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	STANDARED PRICE	20	CN	AUG
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	HIGH PRICE	27	IT	JAN

STEP-5

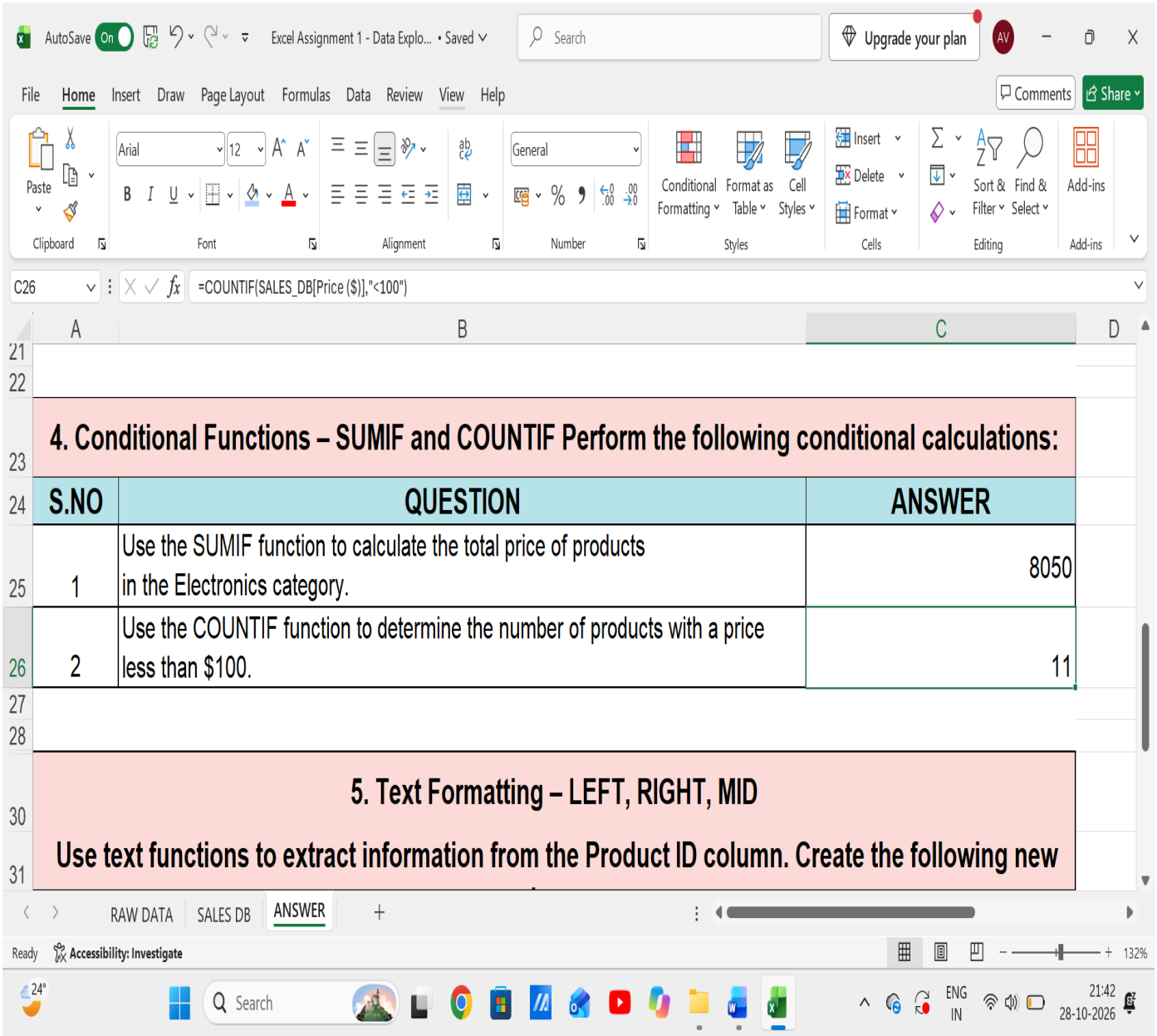
Conditional Functions – SUMIF () and COUNTIF ()

Perform the following conditional calculations:

1.Use the SUMIF function to calculate the total price of products in the Electronics category.



2.Use the COUNTIF function to determine the number of products with a price less than $100.

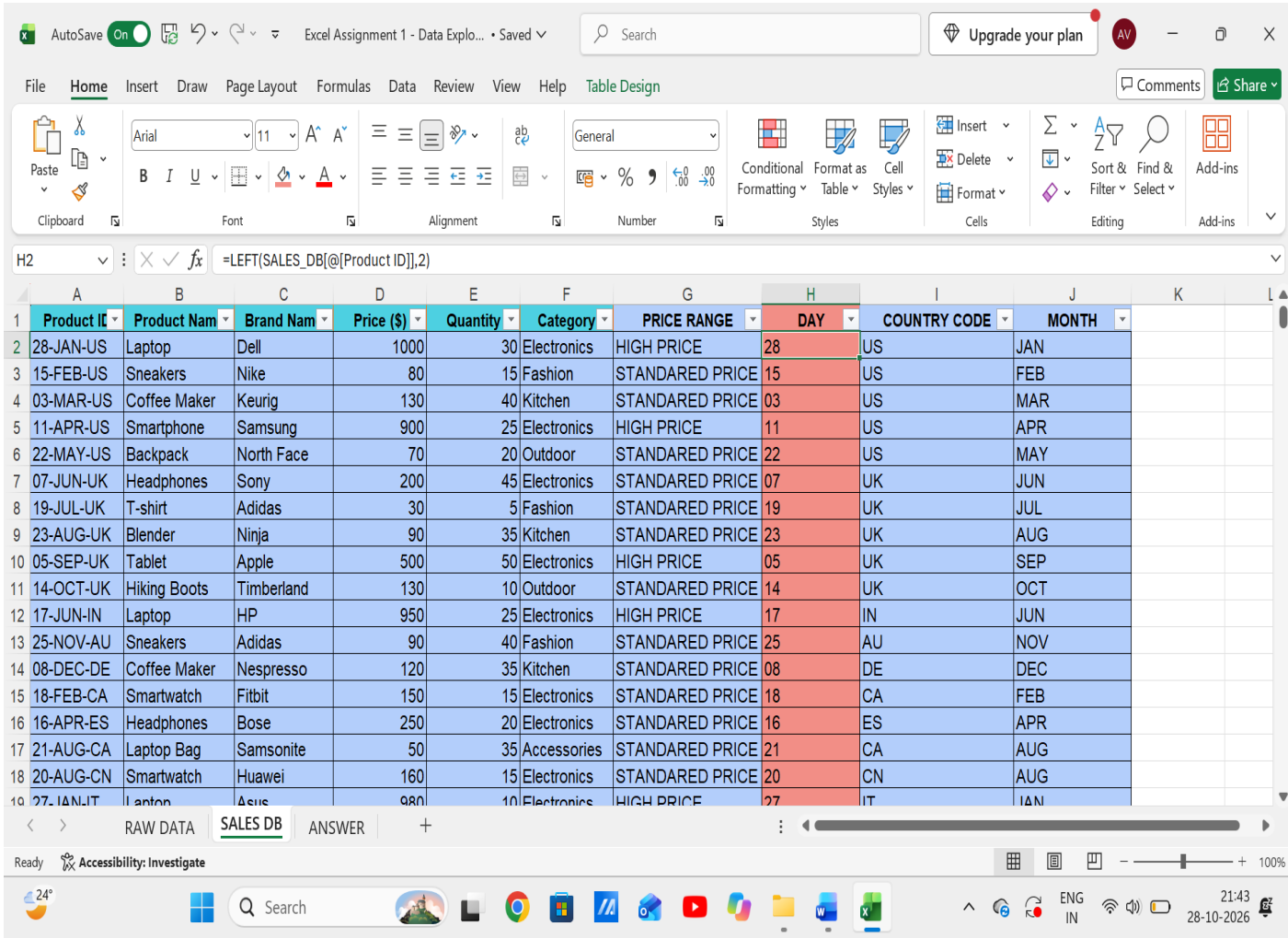


STEP-6

Text Formatting – LEFT, RIGHT, MID

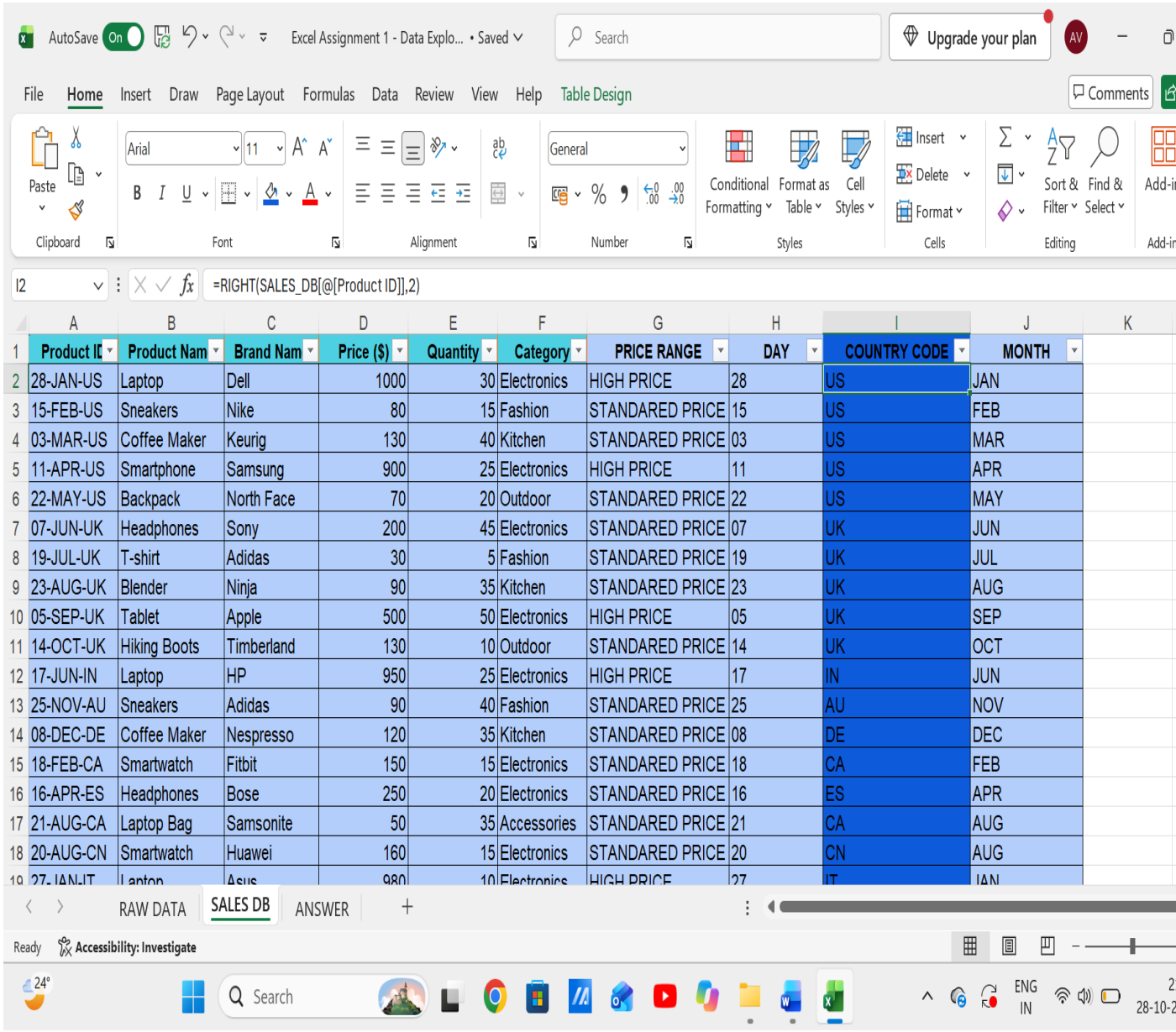
Use text functions to extract information from the Product ID column. Create the following new columns:

1.Create a new column named Day with the first 2 characters of each 'Product ID' using the **LEFT** function.



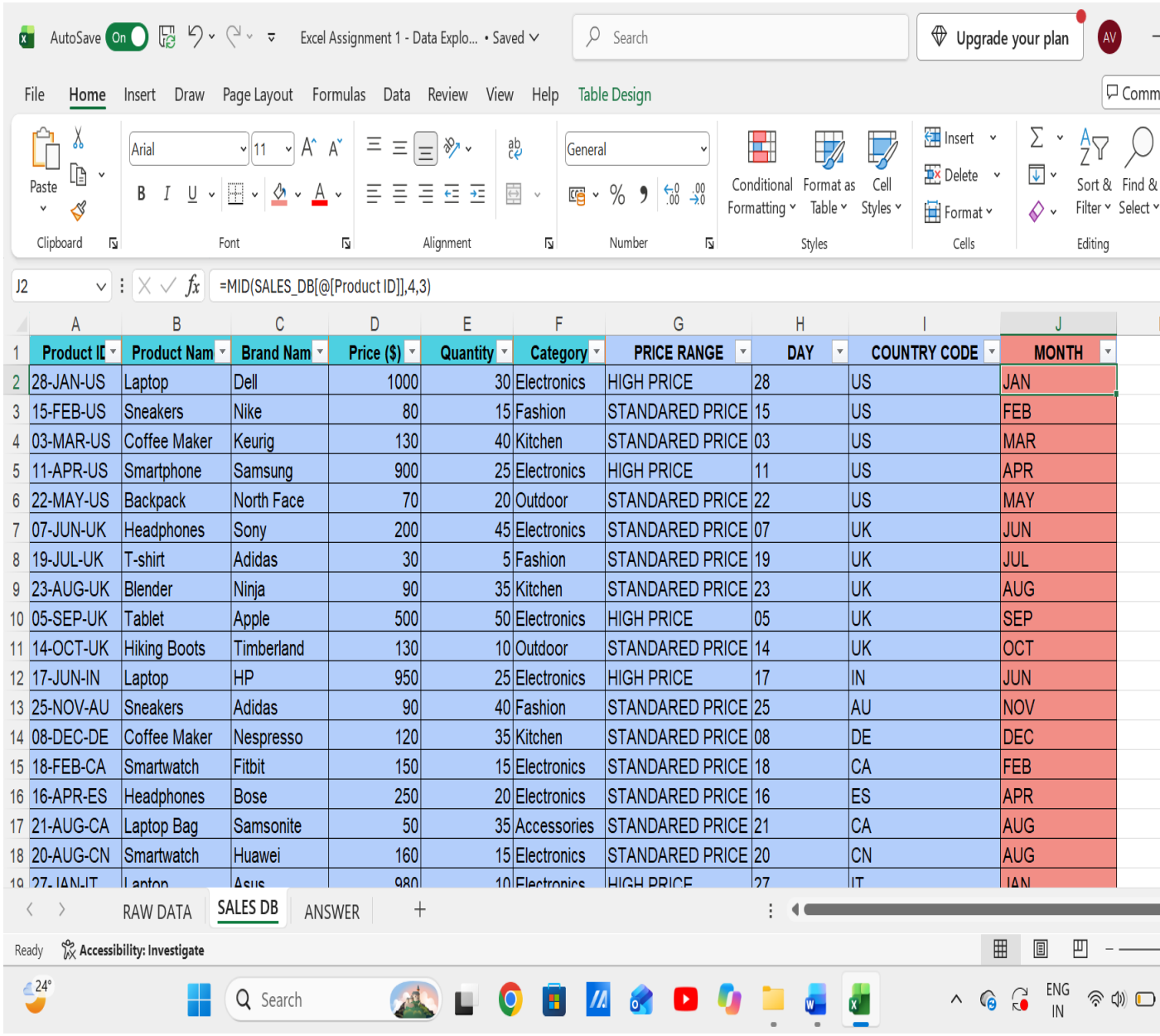
Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category	PRICE RANGE	DAY	COUNTRY CODE	MONTH
28-JAN-US	Laptop	Dell	1000	30	Electronics	HIGH PRICE	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	STANDARED PRICE	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	STANDARED PRICE	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	HIGH PRICE	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	STANDARED PRICE	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	STANDARED PRICE	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	STANDARED PRICE	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	STANDARED PRICE	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	HIGH PRICE	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	STANDARED PRICE	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	HIGH PRICE	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	STANDARED PRICE	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	STANDARED PRICE	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	STANDARED PRICE	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	STANDARED PRICE	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	STANDARED PRICE	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	STANDARED PRICE	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	HIGH PRICE	27	IT	JAN

2.Create a new column named Country Code by extracting the last characters from the 'Product ID' column using the **RIGHT** function.



Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category	PRICE RANGE	DAY	COUNTRY CODE	MONTH
28-JAN-US	Laptop	Dell	1000	30	Electronics	HIGH PRICE	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	STANDARED PRICE	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	STANDARED PRICE	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	HIGH PRICE	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	STANDARED PRICE	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	STANDARED PRICE	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	STANDARED PRICE	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	STANDARED PRICE	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	HIGH PRICE	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	STANDARED PRICE	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	HIGH PRICE	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	STANDARED PRICE	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	STANDARED PRICE	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	STANDARED PRICE	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	STANDARED PRICE	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	STANDARED PRICE	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	STANDARED PRICE	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	HIGH PRICE	27	IT	JAN

3. Create a new column named Month by extracting the 4th to 6th characters from the 'Product ID' column using the **MID** function.



	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category	PRICE RANGE	DAY	COUNTRY CODE	MONTH
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	HIGH PRICE	28	US	JAN
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	STANDARED PRICE	15	US	FEB
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	STANDARED PRICE	03	US	MAR
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	HIGH PRICE	11	US	APR
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	STANDARED PRICE	22	US	MAY
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	STANDARED PRICE	07	UK	JUN
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	STANDARED PRICE	19	UK	JUL
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	STANDARED PRICE	23	UK	AUG
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	HIGH PRICE	05	UK	SEP
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	STANDARED PRICE	14	UK	OCT
12	17-JUN-IN	Laptop	HP	950	25	Electronics	HIGH PRICE	17	IN	JUN
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	STANDARED PRICE	25	AU	NOV
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	STANDARED PRICE	08	DE	DEC
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	STANDARED PRICE	18	CA	FEB
16	16-APR-ES	Headphones	Bose	250	20	Electronics	STANDARED PRICE	16	ES	APR
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	STANDARED PRICE	21	CA	AUG
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	STANDARED PRICE	20	CN	AUG
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	HIGH PRICE	27	IT	JAN